

E1

Consider the circuit in Fig. 1. The circuit parameters at $T = 25^\circ\text{C}$ are the following:

$V_{CC} = 10\text{ V}$; $R_{B1} = 680\text{ k}\Omega$; $R_{B2} = 330\text{ k}\Omega$; $R_{E1} = 1\text{ k}\Omega$; $R_{C1} = 3.3\text{ k}\Omega$; $R_{D2} = 6.8\text{ k}\Omega$; $R_{DE} = 11\text{ k}\Omega$; $R_s = 1\text{ k}\Omega$; $C_s = 100\text{ nF}$; $C_{S2} = 10\text{ }\mu\text{F}$.

Q_1 (NPN BJT): $V_{BE} = 0.7\text{ V}$; $\beta_F = \beta_0 = 200$; $V_{CEsat} = 0.2\text{ V}$; $r_0 = +\infty$;

Q_2 (E-NMOS): $V_{th} = 2\text{ V}$; $I_{DSS} = k_n'(W/2L)V_{th}^2 = 1\text{ mA}$; $r_0 = +\infty$;

$V_T = 26\text{ mV}$.

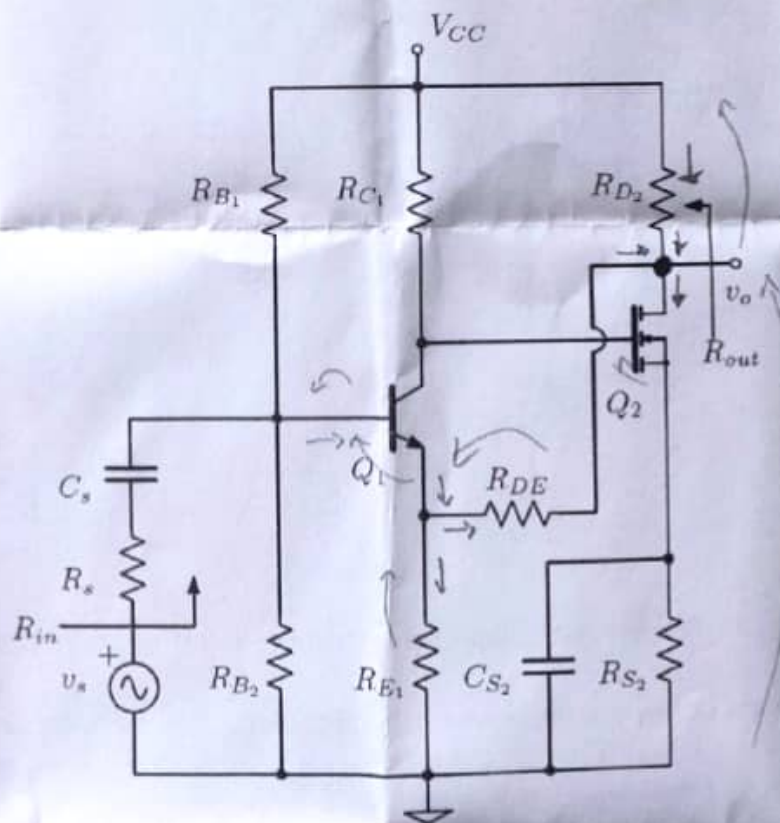


Fig. 1 – A case of two-stage feedback amplifier.

Considering *all capacitors* to be equivalent to *open circuits*, determine:

1. the value of R_{S2} that ensures V_O is biased at 5 V ;
2. the operating points (V_{CE} , I_C) and (V_{DS} , I_D) of Q_1 and Q_2 .

Assuming that, at the frequencies of interest (i.e. at mid-band), capacitors C_s and C_{S2} are equivalent to *short circuits*, determine also:

3. the voltage gain $A_v = v_o/v_s$ of the amplifier;
4. the input resistance R_{in} indicated in the figure;
5. the output resistance R_{out} indicated in the figure.

Finally, considering the effect of capacitors C_s and C_{s2} , calculate also:

6. an estimation of the low frequency bandwidth limit f_L of the amplifier.

E2

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 33 \text{ k}\Omega$; $R_2 = 68 \text{ k}\Omega$; $R_C = 22 \text{ }\Omega$; $C_C = 33 \text{ }\mu\text{F}$.

OpAmp: $A_{d0} = 10^5 \text{ [V/V]}$, $\omega_0 = 100 \text{ rad/s}$, $\omega_1 = 20 \text{ krad/s}$.

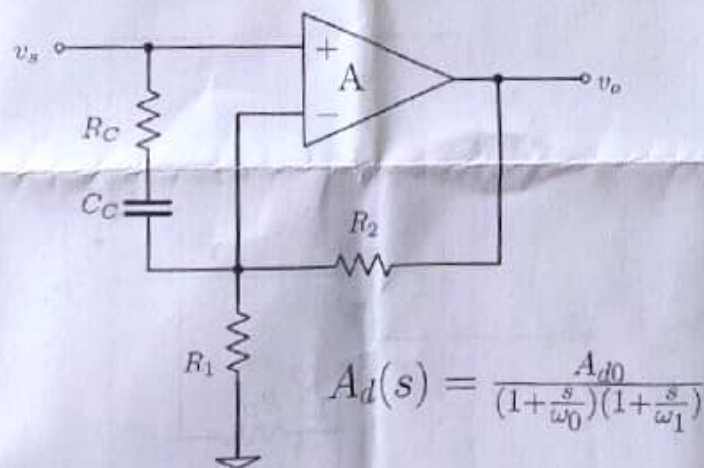


Fig. 2 – Compensated OpAmp in the non-inverting configuration.

Assuming the noise-gain compensation network is disabled, i.e. C_C is open circuited, determine:

1. the expression of the *ideal* circuit voltage gain $A_v(s) = v_o/v_s$;
2. the amplifier bandwidth and *phase margin*.

Then, after activating the compensation network, determine also:

3. the new amplifier bandwidth and phase margin;
4. the expression of the *ideal* circuit voltage gain $A_v(s) = v_o/v_s$.